

2-MINUTE HACKATHON PITCH — **SynBioRem AI**

[Opening — Hook (10–15 sec)]

“Imagine you are managing a polluted river, an oil spill, or toxic industrial waste.
The biggest challenge is not cleaning it — it’s knowing **which biological solution will actually work**.

Right now, scientists rely on slow, expensive trial-and-error methods.

What if we could make that process intelligent?”

[Problem (20 sec)]

“Environmental pollution — from heavy metals, textile dyes, oil spills, and pesticides — is a global crisis.

Biological remediation is promising, but:

- it depends heavily on environmental conditions
- selecting the right microbe is complex
- and there is **no intelligent system** that maps pollutants to optimal biological solutions

This leads to **high cost, long timelines, and inefficient outcomes.**”

[Solution (30 sec)]

“We present **SynBioRem AI** — an AI-assisted platform that recommends and simulates synthetic microbe designs for pollution remediation.

Our system:

- analyzes the pollutant and environmental conditions
- selects the most suitable microbial chassis
- proposes a **synthetic design blueprint**
- and evaluates safety, cost, and scalability

So instead of trial-and-error, we enable **data-driven biological design**.”



[How it works — Interactive (30 sec)]

“Let me show how it works.

A user inputs:

- pollutant type
- pH
- temperature
- environment

Our AI engine:

- profiles the pollutant
- queries a biological knowledge base
- ranks the best candidate microbes

Then, we generate a **synthetic blueprint** and simulate:

- 👉 how effective the solution is
- 👉 how it changes with environmental conditions

Users can adjust parameters in real-time — just like running a virtual experiment.”



[Innovation / X-Factor (20 sec)]

“What makes SynBioRem AI unique is that:

- It combines **AI + synthetic biology + environmental science**
- It is not just a simulator — it is a **decision-making system**
- It introduces **interactive experimentation**, allowing users to explore outcomes dynamically

We are essentially building a **computational assistant for designing biological solutions**.”



[Impact (15 sec)]

“This platform can be used by:

- industries managing waste
- environmental agencies
- and research labs

to **accelerate sustainable remediation strategies at scale.**”



[Closing (10 sec)]

“SynBioRem AI transforms environmental cleanup from guesswork into intelligent design.

- 👉 We don’t just clean pollution —
- 👉 we design the best way to eliminate it.”



DELIVERY TIPS (IMPORTANT)

- Speak **slow + confident**
- Use hand gestures when saying:
 - “User inputs...”
 - “AI recommends...”
- Pause after the hook
- Smile slightly at closing line



OPTIONAL POWER LINE (if time allows)

Add at the end:

“This is not just a project — it’s a step toward programmable environmental solutions.”